

# Negative results and claim revision

Project: Reproducible AI Research Study

## 1. Your mission

### Your learning target

I can explain the mechanism and terminology of negative results and claim revision.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Robustness across conditions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Running the primary experiment. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Pilot experiments. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Explain the mechanism and terminology of negative results and claim revision

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

### Retrieve what you already know

Recall Robustness across conditions. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

### My starting explanation

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### This week's end goal

Explain negative results and claim revision using the records and boundaries of this fictional investigation.

## 2. Notice the evidence

Negative results report that the evidence did not meet a declared expectation. They can reveal design limits or useful constraints without proving no effect anywhere.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

**List the given facts and one fact that is not supplied.**

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## 3. See how it works

Compare results with the original criterion and uncertainty assumptions. Preserve the failed hypothesis, discuss plausible mechanisms and label revised questions as new work.

### Negative results and claim revision — evidence trace

**Input and scope · Check or calculation · Conclusion and limit**

**Input and scope:** Hypothesis expects B support strictly above A. Both pass 8/10 reserved cases.

**Check or calculation:** Compare results with the original criterion and uncertainty assumptions. Preserve the failed hypothesis, discuss plausible mechanisms and label revised questions as new work.

**Conclusion and limit:** Strict improvement is not observed; both 80%. This does not prove population equality or universal uselessness.

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The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for

qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

## Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

## 4. Compare the cases

### Compare a familiar case with a changed case

Hypothesis expects B support strictly above A. Both pass 8/10 reserved cases. Show the intermediate steps and give the conclusion with its scope.

**Keep the procedure fixed. Explain what changed in the inputs and result.**

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### Change one thing in the pictured case

Change: B wins on one new task after the team chooses it because A fails.

Prediction: Treat that selected demonstration as exploratory, not a replacement for the original result.

Reason: Case selection now depends on outcomes.

## 5. Words that unlock the idea

### Word · Meaning

**Word:** Negative result

**Meaning:** Evidence failing to meet the declared expectation

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**Word:** Claim revision

**Meaning:** A bounded change to the conclusion

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**Word:** Equivalence

**Meaning:** A separate claim requiring its own design and criterion

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**Use two terms to explain a step in this week's example.**

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## 6. Follow a worked example

Hypothesis expects B support strictly above A. Both pass 8/10 reserved cases.

Strict improvement is not observed; both 80%. This does not prove population equality or universal uselessness.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

### Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

### A pictorial worked case: Negative results and claim revision

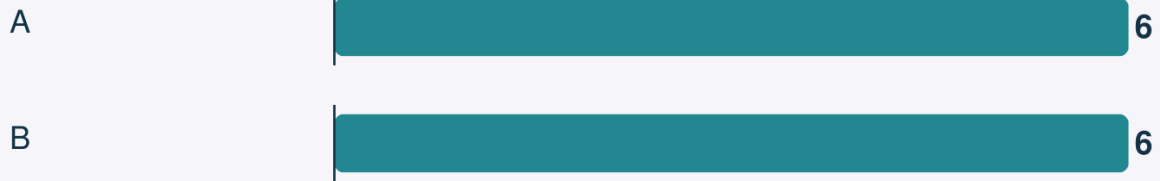
A negative result changes the claim. A proposed prompt improvement ties the baseline on a fixed eight-case evaluation.

### The facts we will use

A and B each score 6/8. Before running, the team predicted B would score higher.

## No observed count advantage

correct cases out of 8 shared cases



Bar length shows the supplied value; the marked baseline is zero.

## Negative results and claim revision

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



### 1. Recall the prediction

The claim was a higher correct count for B.



### 2. State the finding

Both versions score 75%; no observed count advantage appears here.



### 3. Revise cautiously

Report the unsupported prediction and a justified next test.

### CHANGE ONE THING

B wins on one new task after the team chooses it because A fails.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

### Step · What happens and why

**Step:** 1. Recall the prediction

**What happens and why:** The claim was a higher correct count for B.

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**Step:** 2. State the finding

**What happens and why:** Both versions score 75%; no observed count advantage appears here.

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### Step: 3. Revise cautiously

**What happens and why:** Report the unsupported prediction and a justified next test.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

A negative result can be informative when methods and limits are clear. A tie does not prove the methods are equivalent under all conditions or that the experiment was worthless.

### What this example does not establish

A tie on eight cases is not an equivalence test; uncertainty and practical thresholds remain unspecified.

Read the labelled picture: Does a 6/8 tie support the prediction that B scores higher? Explain why one newly selected favourable demonstration does not replace the original result or prove equivalence.

Change one thing: B wins on one new task after the team chooses it because A fails. Predict the result before checking the explanation. What remains the same?

## 7. Find and repair a mistake

Failure to meet a criterion or reject a null is not proof of exact equivalence.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

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### Locate the earliest unsupported step

A report omits this qualification: Failure to meet a criterion or reject a null is not proof of exact equivalence. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**State one repair and the checks you would rerun.**

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## 8. Practise together

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

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**Explain your prediction to a partner before comparing results.**

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### Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Negative results and claim revision”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

**Plan · Do · Check**

**Plan:** What is given? Which rule or procedure applies?

**Do:** Record each intermediate step.

**Check:** Does the result follow? What remains unknown?

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Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

### Check your reasoning

Use the worked case for Negative results and claim revision. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

### Explain the picture

Does a 6/8 tie support the prediction that B scores higher? Explain why one newly selected favourable demonstration does not replace the original result or prove equivalence.

#### My explanation using a pictured step and the given facts

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This is supported practice. Keep the separately named independent case for an attempt before feedback.

## 9. Run the investigation

### Predict, investigate and preserve the evidence

Hypothesis expects B support strictly above A. Both pass 8/10 reserved cases. Show the intermediate steps and give the conclusion with its scope.

#### Record · Your evidence

**Record:** Given input and fixed rule

**Your evidence:** \_\_\_\_\_

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**Record:** Prediction before checking

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Actual result, or “not run”

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** First mismatch and one explanation

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Boundary or missing input and expected response

**Your evidence:** \_\_\_\_\_

\_\_\_\_\_

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

### Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

## 10. Build your understanding

### Explain → apply → challenge

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

A report omits this qualification: Failure to meet a criterion or reject a null is not proof of exact equivalence. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

**Explain which evidence supports your conclusion and which claim remains unproved.**

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## 11. Connect your project

Add a labelled negative results and claim revision evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of negative results and claim revision. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

### Project evidence · Record

**Project evidence:** Component and version

**Record:** \_\_\_\_\_

\_\_\_\_\_

**Project evidence:** Inputs permitted at prediction time

**Record:** \_\_\_\_\_

\_\_\_\_\_

**Project evidence:** Trace and comparison with earlier work

**Record:** \_\_\_\_\_

\_\_\_\_\_

**Project evidence:** Failure, limit and human review

**Record:** \_\_\_\_\_

\_\_\_\_\_

### Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

## 12. Show what you know

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

**Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

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For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

**My limited claim and a useful next check**

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## 13. Study a complete case

### Complete evidence record

### Negative results and claim revision — evidence trace

**Input and scope · Check or calculation · Conclusion and limit**

**Input and scope:** Hypothesis expects B support strictly above A. Both pass 8/10 reserved cases.

**Check or calculation:** Compare results with the original criterion and uncertainty assumptions. Preserve the failed hypothesis, discuss plausible mechanisms and label revised questions as new work.

**Conclusion and limit:** Strict improvement is not observed; both 80%. This does not prove population equality or universal uselessness.

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The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain

every intermediate step. The text/table accompanies the figure so the example can be read without colour.

**My complete trace of the worked case**

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# 14. Try a new case independently

**Independent case: Apply a new case independently**

Hypothesis predicts mean latency reduction at least 2 seconds. Observed paired mean reduction .8. Report.

**My independent explanation, trace and limits**

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

**Attempt this without hints**

Hypothesis predicts mean latency reduction at least 2 seconds. Observed paired mean reduction .8. Report.

**My independent procedure, result and limitation**

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Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

**Use feedback to improve**

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

**15. Your lesson route**

Start with this question: A negative result changes the claim. A proposed prompt improvement ties the baseline on a fixed eight-case evaluation.

**By the end, I can · How I will show it**

**By the end, I can:** Explain the mechanism and terminology of negative results and claim revision

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Apply the stated procedure to a new case with a traceable result

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Identify an assumption or limitation and design a revealing follow-up

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

## 16. Finish and return

Close your notes. Explain this goal in your own words: Explain the mechanism and terminology of negative results and claim revision

### My deciding step, evidence and one remaining question

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If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

## 17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A negative result changes the claim. A proposed prompt improvement ties the baseline on a fixed eight-case evaluation.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Report the unsupported prediction and a justified next test.

1. Revise cautiously
2. Recall the prediction
3. State the finding

### My selected conclusion, deciding fact and explanation

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Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Revise cautiously Report the unsupported prediction and a justified next test. A negative result can be informative when methods and limits are clear. A tie does not prove the methods are equivalent under all conditions or that the experiment was worthless.

## Practice 2: explain the deciding evidence

Changed-case practice: B wins on one new task after the team chooses it because A fails. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. Treat that selected demonstration as exploratory, not a replacement for the original result.

### My selected conclusion, deciding fact and explanation

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Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Treat that selected demonstration as exploratory, not a replacement for the original result. Case selection now depends on outcomes.

## Repair a missing building block

Review Robustness across conditions (ai-7/27). Goal: Explain the mechanism and terminology of robustness across conditions. Short practice: Calculate indoor, outdoor and combined rates. Which relevant weakness disappears from view when only the combined rate is reported?. Return to this week and explain what you changed.

Review Running the primary experiment (ai-7/25). Goal: Explain the mechanism and terminology of running the primary experiment. Short practice: Compute A's and B's six-case counts and identify the cases that favour each. How must a B revision tuned on case four be labelled when evaluated again on the same set?. Return to this week and explain what you changed.

Review Pilot experiments (ai-7/22). Goal: Explain the mechanism and terminology of pilot experiments. Short practice: Calculate reviewer agreement on the four pilot answers. What can the disagreement improve, and why are those revised practice answers not untouched final evidence?. Return to this week

and explain what you changed.

## **Keep your completed work**

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.